



Candidate Number \_\_\_\_\_

## **GOSFORD HIGH SCHOOL**

**2007**

**TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION**

## **MATHEMATICS EXTENSION 2**

### **General Instructions:**

- Reading time – 5minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- Each question should be started on a new page.
- All necessary working should be shown in every question

**Total marks: - 120**

- Attempt Questions 1 -8
- All questions are of equal value

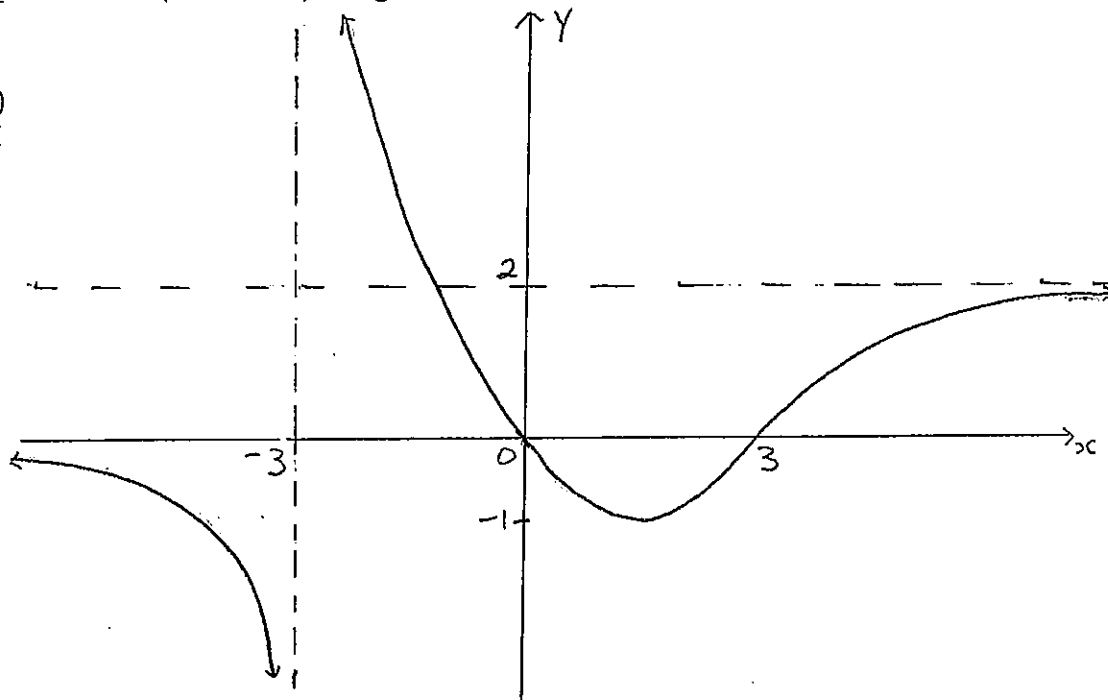
**Question 1 (15 Marks)****Marks**

- a) Find  $\int \frac{x}{\sqrt{9-4x^2}} dx$  2
- b) Find  $\int_1^e x^5 \log_e x \, dx$  3
- c) (i) Find real numbers  $a$ ,  $b$  and  $c$  such that 2
- $$\frac{8}{(x+2)(x^2+4)} = \frac{a}{x+2} + \frac{bx+c}{x^2+4}$$
- (ii) Hence show that  $\int_0^2 \frac{8dx}{(x+2)(x^2+4)} = \frac{1}{2} \log 2 + \frac{\pi}{4}$  4
- d) Use the substitution  $x = 3 \sin \theta$  to evaluate  $\int_0^{\frac{1}{\sqrt{2}}} \frac{dx}{(9-x^2)^{\frac{3}{2}}}$  4

**Question 2 (15 Marks) Begin a New Booklet**

- a) The zeros of  $(x-1)(x+i)$  are obviously 1 and  $-i$ . These are not complex conjugates. How do you explain this? 2
- b) Let  $\alpha = -1 + i\sqrt{3}$
- (i) Find the exact value of  $|\alpha|$  and  $\arg \alpha$
- (ii) Find the exact value of  $\alpha^7$  in the form  $a+ib$  where  $a$  and  $b$  are real. 4
- c) Find the square roots of  $-5-12i$  in the form  $a+ib$ . 4
- d) The equation  $|z-1-3i|+|z-9-3i|=10$  corresponds to an ellipse in the Argand diagram.
- (i) Write down the complex number corresponding to the centre of the ellipse. 1
- (ii) Sketch the ellipse, and state the lengths of the major and minor axes. 3
- (iii) Write down the range of values of  $\arg(z)$  for complex numbers  $z$  corresponding to points on the ellipse. 1

a)



The diagram shows the graph of  $y = f(x)$ .

Draw separate half page sketches of:

(i)  $y = (f(x))^2$  2

(ii)  $y = \sqrt{f(x)}$  2

(iii)  $y^2 = f(x)$  1

(iv)  $y = \frac{1}{f(x)}$  2

(v)  $y = xf(x)$  2

b) (i) Prove that the tangent at a point  $(x_1, y_1)$  to  $xy = c^2$  is  $xy_1 + x_1y = 2c^2$  2

(ii) P is a point of intersection of the rectangular hyperbolas  $x^2 - y^2 = a^2$  and  $xy = c^2$ .

The tangent at P to the first hyperbola meets its asymptotes in A and C, and The tangent at P to the second hyperbola meets its asymptotes in B and D. Prove that ABCD is a square. 4

**Question 4 (15 Marks) Begin a New Booklet**

- a) A particle P of mass  $m$  moves with constant angular velocity  $\omega$  on a circle of radius  $r$ . Its position at time  $t$  is given by:

$$\begin{aligned}x &= r \cos \theta \\y &= r \sin \theta\end{aligned}\quad \text{where } \theta = \omega t.$$

- (i) Show that there is an inward radial force of magnitude  $mr\omega^2$  acting on P. 3
- (ii) A telecommunications satellite, of mass  $m$ , orbits Earth with constant angular velocity  $\omega$  at a distance  $r$  from the centre of Earth. The gravitational force exerted by Earth on the satellite is  $\frac{Am}{r^2}$  where  $A$  is a constant.

By considering all other forces on the satellite to be negligible, show that

$$r = \sqrt[3]{\frac{A}{\omega^2}} \quad \text{1}$$

- b) It is given that  $x, y, z$  are positive numbers. Prove that

- (i)  $x^2 + y^2 \geq 2xy$  1
- (ii)  $x^2 + y^2 + z^2 - xy - yz - zx \geq 0$  2

Multiply both sides of the inequality in (ii) by  $(x+y+z)$  to show

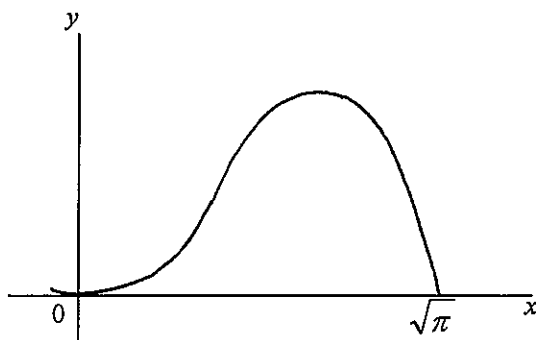
- (iii)  $x^3 + y^3 + z^3 \geq 3xyz$  2

Deduce from (iii) or prove otherwise, that

- (iv)  $(x+y+z)(x^{-1} + y^{-1} + z^{-1}) \geq 9$  2

- c) The curve  $y = \sin(x^2)$  from  $x=0$  to  $x=\sqrt{\pi}$  is rotated about the  $y$ -axis. 4

Sketch a typical cylindrical shell and use this method to find the volume formed.



**Question 5 (15 Marks) Begin a New Booklet**

- a) Prove that if two polynomials  $P(x)$  and  $Q(x)$  have a common factor of  $(x-a)$ , then  $(x-a)$  is also a factor of  $P(x) - Q(x)$ . 2

Hence find the value of  $k$  if  $x^3 + x^2 - 5x + k$  and  $x^3 - 8x^2 + 13x - 2k$  have a common double root. What is this double root? 3

- b) (i) Find the expansion of  $\cos 5\theta$  and  $\sin 5\theta$  in terms of powers of  $\cos \theta$  and  $\sin \theta$ . 4
- (ii) Use the results of (i) to obtain an expression for  $\tan 5\theta$  in terms of powers of  $\cos \theta$  and  $\sin \theta$ . Hence prove

$$\tan 5\theta = \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta} \quad 3$$

- (iii) Hence solve the equation  $t^5 - 5t^4 - 10t^3 + 10t^2 + 5t - 1 = 0$  3

**Question 6 (15 Marks) Begin a New Booklet**

- a) A particle of mass  $m$  falls vertically from rest, from a point  $O$ , in a medium whose resistance is  $mkv$ , where  $k$  is a positive constant and  $v$  its velocity.
- (i) Obtain an expression for its velocity after  $t$  seconds. 3

An equal particle is projected vertically upwards from  $O$  with initial velocity  $u$ , in the same medium. This particle is released simultaneously with the first particle.

- (ii) Show that the velocity of the first particle when the second particle is momentarily at rest, is given by  $\frac{Vu}{V+u}$  where  $V$  is the terminal velocity of the first particle. 5

- b) A solid has as its base the circle  $x^2 + y^2 = a^2$  in the  $xy$  plane. Find the volume of the solid such that every cross-section by a plane parallel to the  $x$ -axis is a square with one side in the  $xy$ -plane. 4

- c) If  $m$  and  $n$  are positive integers and  $m \neq n$ , show that

$$\int_0^\pi \cos mx \cos nx dx = 0 \quad 3$$

**Question 7 (15 Marks) Begin a New Booklet**

- a) The curves  $y = \cos x$  and  $y = \tan x$  intersect at a point P whose x coordinate is  $\alpha$ .  
 (i) Show that the curves intersect at right angles at P

3

(ii) Show that  $\sec^2 \alpha = \frac{1 + \sqrt{5}}{2}$

2

b) If  $U_n = \int_0^1 (1-x^2)^n dx$  show that  $U_n = \frac{2n}{2n+1} U_{n-1}$

4

Hence evaluate  $U_4$

1

- c) In the diagram, ABC is a triangle inscribed in a circle. P is a point on the minor arc AB.  
 L, M, and N are the feet of the perpendiculars from P to CA (produced), AB and BC respectively.

- (i) State a reason why P, M, A and L are concyclic points.

1

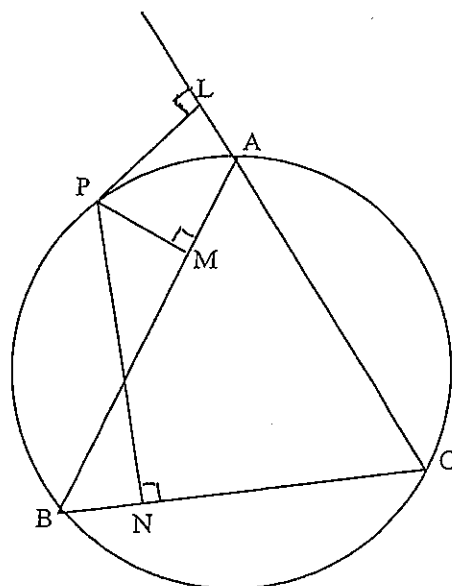
- (ii) State a reason why P, B, N and M are concyclic points.

1

- (iii) Join BP, PA, LM and MN.

Use the three cyclic quadrilaterals to prove that L, M and N are collinear.

Hint: Let  $\angle PBN = \alpha$



3

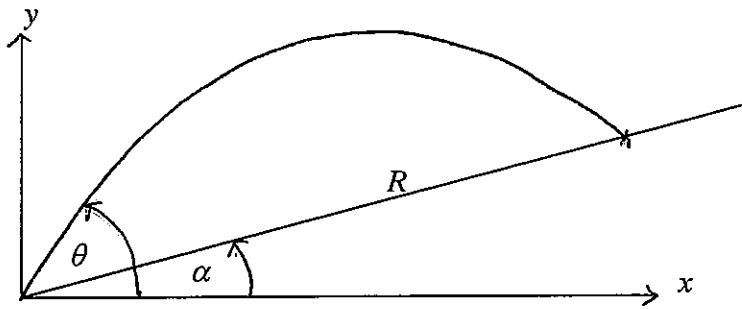
**Question 8 (15 Marks) Begin a New Booklet**

- a) From a diagram, show that  $\sin x < x < \tan x$  if  $0 < x < \frac{\pi}{2}$ .

Hence prove that  $\int_0^{\frac{\pi}{6}} x^2 \sin x \, dx < \frac{\pi^4}{2^6 \cdot 3^4} < \int_0^{\frac{\pi}{6}} x^2 \tan x \, dx.$  3

- b) Solve for  $x$   $\tan^{-1} 3x - \tan^{-1} 2x = \tan^{-1} \frac{1}{5}$  3

- c) A projectile is fired with velocity  $V$ , at an angle  $\theta$  to the horizontal, up a plane inclined at an angle  $\alpha$  to the horizontal, (where  $\alpha$  and  $V$  are constants).



Neglecting air resistance, and using  $g$  for acceleration due to gravity,

- (i) Write down the equations of motion for the projectile. 1

- (ii) Show that the equation of the trajectory is

$$y = -\frac{g}{2V^2} x^2 \sec^2 \theta + x \tan \theta$$
 2

- (iii) Write down the equation of the inclined plane.

By solving simultaneously for  $x$ , and noting that  $R = x \sec \alpha$ , 2

find an expression for the range  $R$  on the inclined plane.

- (v) Hence show that  $\frac{dR}{d\theta} = \frac{2V^2}{g} (\cos 2\theta + \sin 2\theta \tan \alpha) \sec \alpha$  2

- (vi) Find the value of  $\theta$  (in terms of  $\alpha$ ) which gives the maximum range. 2

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE:

$$\ln x = \log_e x, x > 0$$

## Solutions to 2007 Trial HSC Extension 2

### Question 1

$$\begin{aligned} \text{a)} \quad \int \frac{x}{\sqrt{9-4x^2}} dx &= -\frac{1}{8} \int -8x (9-4x^2)^{-\frac{1}{2}} dx \\ &= -\frac{1}{8} \cdot 2 (9-4x^2)^{\frac{1}{2}} = -\frac{1}{4} \sqrt{9-4x^2} + C \end{aligned}$$

$$\begin{aligned} \text{b)} \quad \int_1^e x^5 \ln x \, dx & \quad u = \ln x \quad dv = x^5 \, dx \\ & \quad du = \frac{1}{x} \, dx \quad v = \frac{x^6}{6} \\ \int_1^e x^5 \ln x \, dx &= \left[ \frac{x^6}{6} \ln x \right]_1^e - \int_1^e \frac{x^6}{6} \cdot \frac{1}{x} \, dx \\ &= \left[ \frac{x^6}{6} \ln x \right]_1^e - \frac{1}{6} \int_1^e x^5 \, dx \\ &= \left[ \frac{x^6}{6} \ln x - \frac{x^6}{36} \right]_1^e \\ &= \left( \frac{e^6}{6} - \frac{e^6}{36} \right) - \left( 0 - \frac{1}{36} \right) = \underline{\underline{\frac{5e^6 - 1}{36}}} \end{aligned}$$

$$\begin{aligned} \text{c) i)} \quad \frac{8}{(x+2)(x^2+4)} &= \frac{a}{x+2} + \frac{bx+c}{x^2+4} \\ &= \frac{a(x^2+4) + (x+2)(bx+c)}{(x+2)(x^2+4)} \\ \therefore 8 &\equiv a(x^2+4) + (x+2)(bx+c) \end{aligned}$$

$$\text{Sub } x = -2: \quad 8 = 8a \implies a = 1$$

$$\text{Equate coeffs of } x^2 \quad 0 = a + b \implies b = -1$$

$$\text{Sub } x = 0$$

$$8 = 4a + 2c$$

$$4 = 2a + c \implies c = 2$$

$$\begin{aligned} \text{ii)} \quad \int_0^2 \frac{1}{x+2} + \frac{-x+2}{x^2+4} \, dx \\ = \int_0^2 \frac{1}{x+2} - \frac{1}{2} \cdot \frac{2x}{x^2+4} + \frac{2}{x^2+4} \, dx \end{aligned}$$

$$\begin{aligned}
 \text{i) cont} &= \left[ \ln(x+2) - \frac{1}{2} \ln(x^2+4) + 2 \cdot \frac{1}{2} \tan^{-1} \frac{x}{2} \right]_0^2 \\
 &= \left( \ln 4 - \frac{1}{2} \ln 8 + \tan^{-1} 1 \right) - \left( \ln 2 - \frac{1}{2} \ln 4 + \tan^{-1} 0 \right) \\
 &= 2 \ln 2 - \frac{3}{2} \ln 2 + \frac{\pi}{4} - \ln 2 + \ln 2 - 0 \\
 &= \frac{1}{2} \ln 2 + \frac{\pi}{4} \quad \text{as required.}
 \end{aligned}$$

$$\begin{aligned}
 \text{d)} \quad &\int_0^{\frac{3}{\sqrt{2}}} \frac{dx}{(9-x^2)^{3/2}} \qquad \text{Put } x = 3 \sin \theta \quad \text{When } x=0, \theta=0 \\
 &\qquad \qquad \qquad dx = 3 \cos \theta d\theta \quad \text{When } x = \frac{3}{\sqrt{2}}, \theta = \frac{\pi}{4}
 \end{aligned}$$

$$= \int_0^{\frac{\pi}{4}} \frac{3 \cos \theta d\theta}{(9-9 \sin^2 \theta)^{3/2}}$$

$$= \int_0^{\frac{\pi}{4}} \frac{3 \cos \theta d\theta}{(9 \cos^2 \theta)^{3/2}}$$

$$= \int_0^{\frac{\pi}{4}} \frac{3 \cos \theta d\theta}{27 \cos^3 \theta}$$

$$= \int_0^{\frac{\pi}{4}} \frac{d\theta}{9 \cos^2 \theta}$$

$$= \frac{1}{9} \int_0^{\frac{\pi}{4}} \sec^2 \theta d\theta$$

$$= \frac{1}{9} \left[ \tan \theta \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{9} \left( \tan \frac{\pi}{4} - \tan 0 \right)$$

$$= \frac{1}{9}$$

## Question 2

$$a) (x-1)(x+i) = x^2 + (i-1)x - i$$

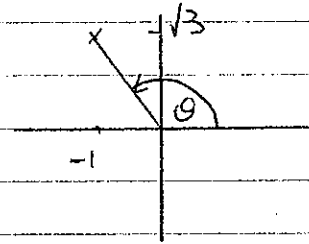
$$P(x) = x^2 + (i-1)x - i$$

The coefficients of  $P(x)$  are not all real. Therefore the zeros are not in conjugate pairs.

$$b) i) \alpha = -1 + i\sqrt{3}$$

$$|\alpha|^2 = 1^2 + 3$$

$$|\alpha| = 2$$



$$\theta = \arg \alpha = \pi + \tan^{-1} \sqrt{3}$$
$$= \frac{2\pi}{3}$$

$$ii) -1 + i\sqrt{3} = 2 \left( \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$$

$$\alpha^n = 2^n \left( \cos \frac{2\pi}{3} \right)^n$$

$$= 128 \cos \frac{14\pi}{3}$$

$$\text{But } \frac{14\pi}{3} \equiv \frac{2\pi}{3}$$

$$= 128 \cos \frac{2\pi}{3}$$

$$= 128 (-1 + i\sqrt{3})$$

$$= -128 + 128i\sqrt{3}$$

$$\therefore a = -128, \quad b = 128\sqrt{3}$$

$$c) z^2 = -5 - 12i = (a + ib)^2$$

$$a^2 - b^2 = -5$$

$$2ab = -12 \Rightarrow ab = -6$$

$$a^2 - \frac{36}{a^2} = -5$$

$$a^4 + 5a^2 - 36 = 0$$

$$(a^2 + 9)(a^2 - 4) = 0$$

$$a^2 = -9 \quad \text{or} \quad 4$$

$$a = \pm 2 \quad (\text{real value})$$

$$b = -\frac{6}{a} = \mp \frac{6}{2} = \mp 3$$

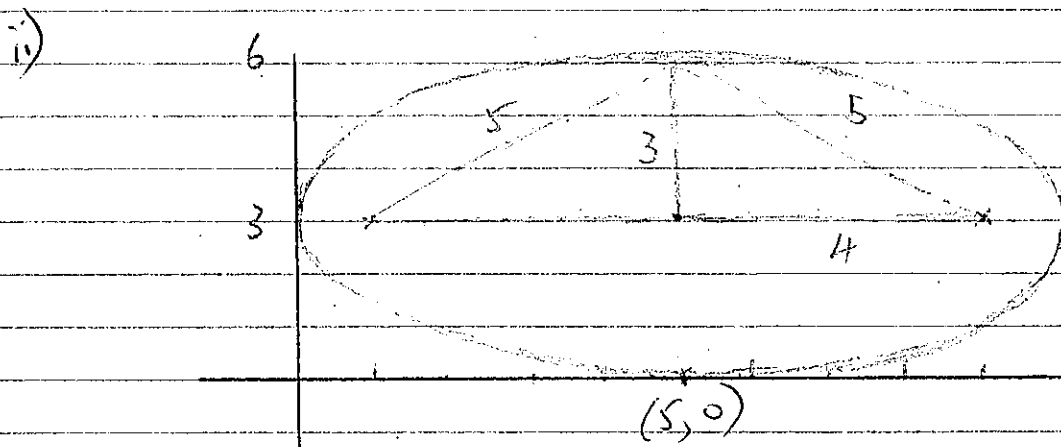
$$\therefore \sqrt{-5-12i} = 2-3i \quad \text{or} \quad -2+3i$$

$$d) \quad |z - 1 - 3i| + |z - 9 - 3i| = 10$$

$$|z - (1+3i)| + |z - (9+3i)| = 10$$

Distance of  $z$  from  $(1+3i)$  + distance from  $(9+3i)$   
 $= 10$

i) Centre of ellipse is mid point  
 ie  $(5+3i)$



Major axis 10 units

Minor axis 6 units

iii) At  $(5, 0)$   $\arg z = 0$

At  $(0, 3)$   $\arg z = \frac{\pi}{2}$

$$0 \leq \arg z \leq \frac{\pi}{2}$$

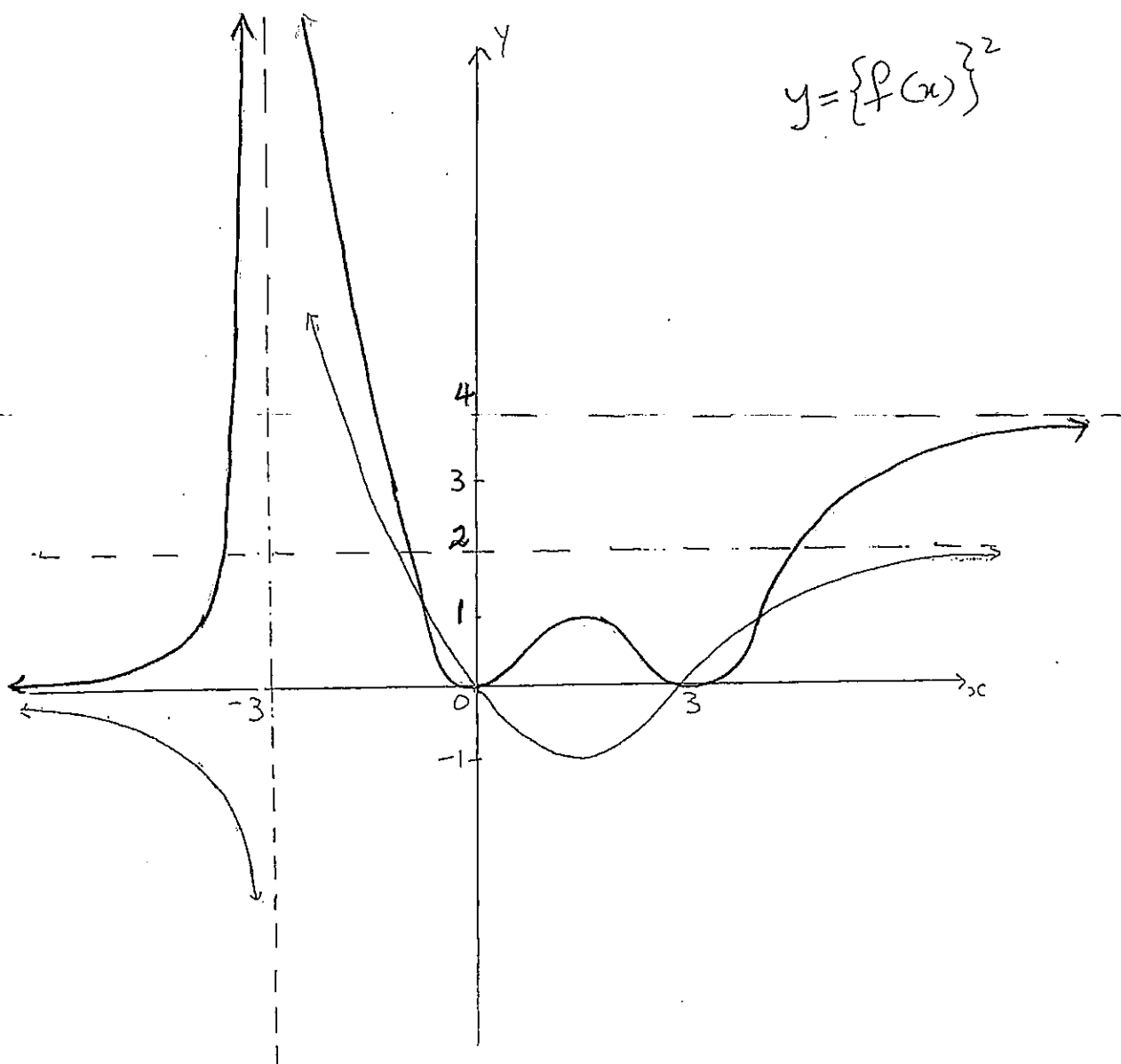
(Note equation of ellipse is

$$\frac{(x-5)^2}{25} + \frac{(y-3)^2}{9} = 1$$

### Question 3

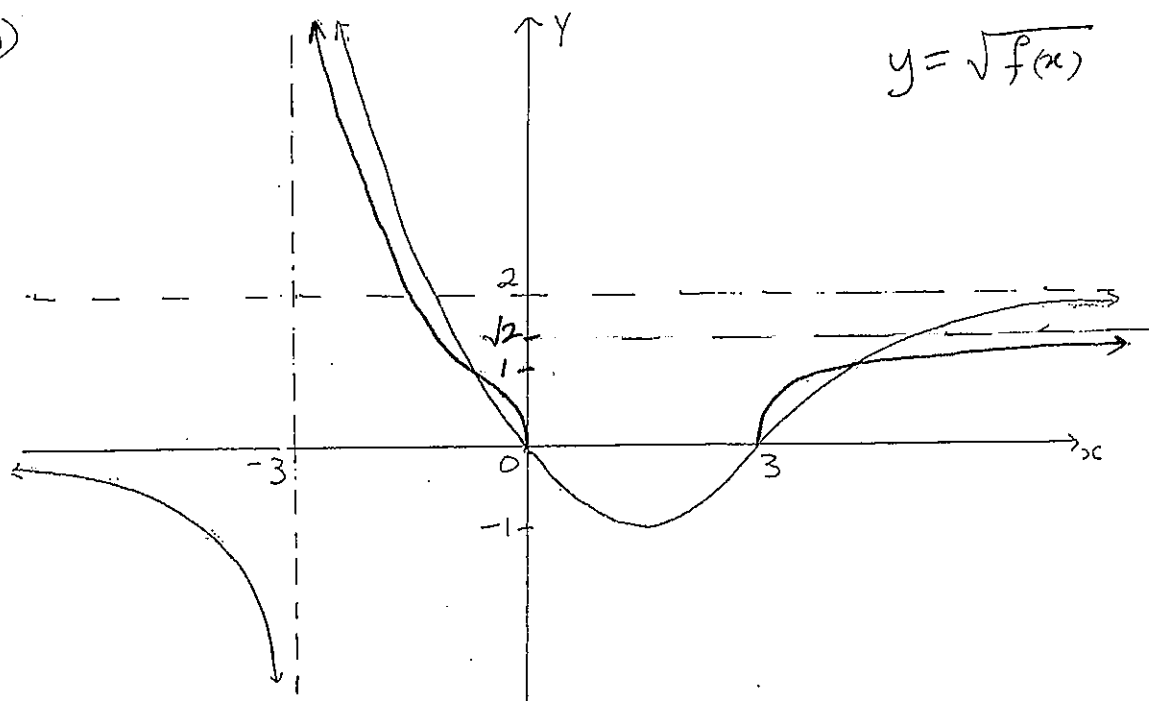
a)

i)



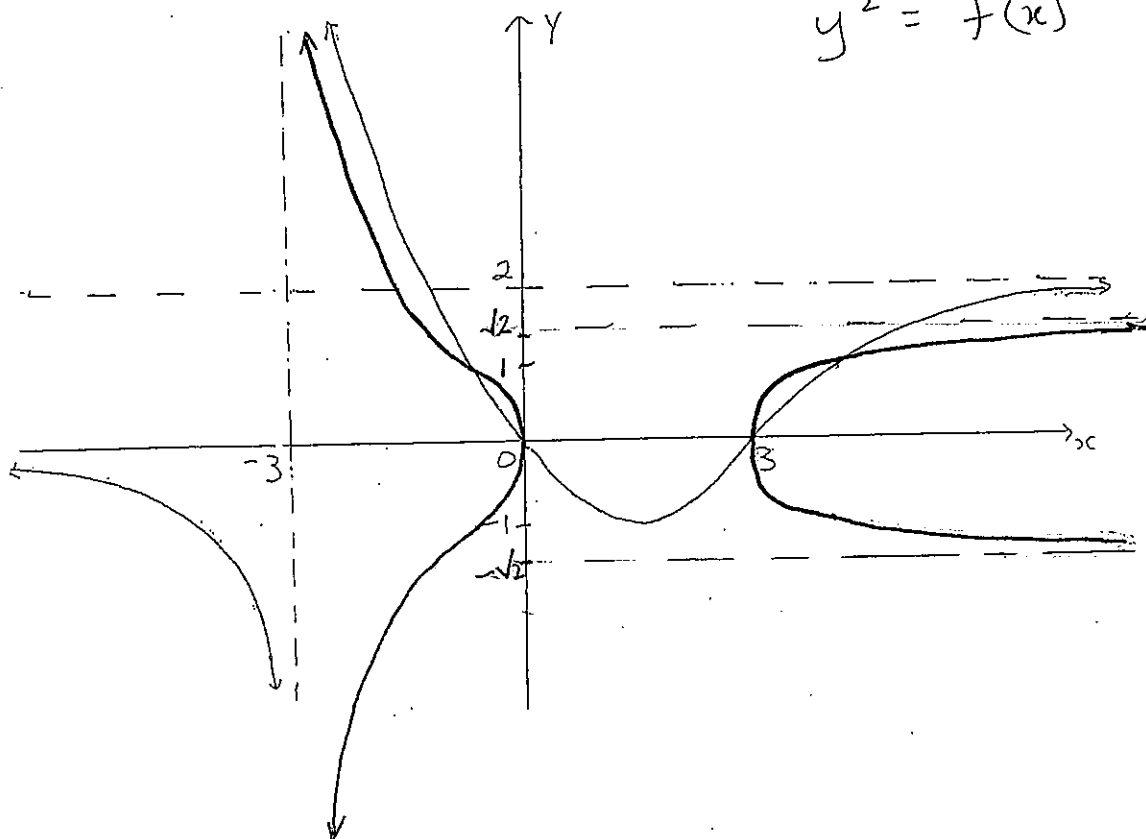
i)

$$y = \sqrt{f(x)}$$



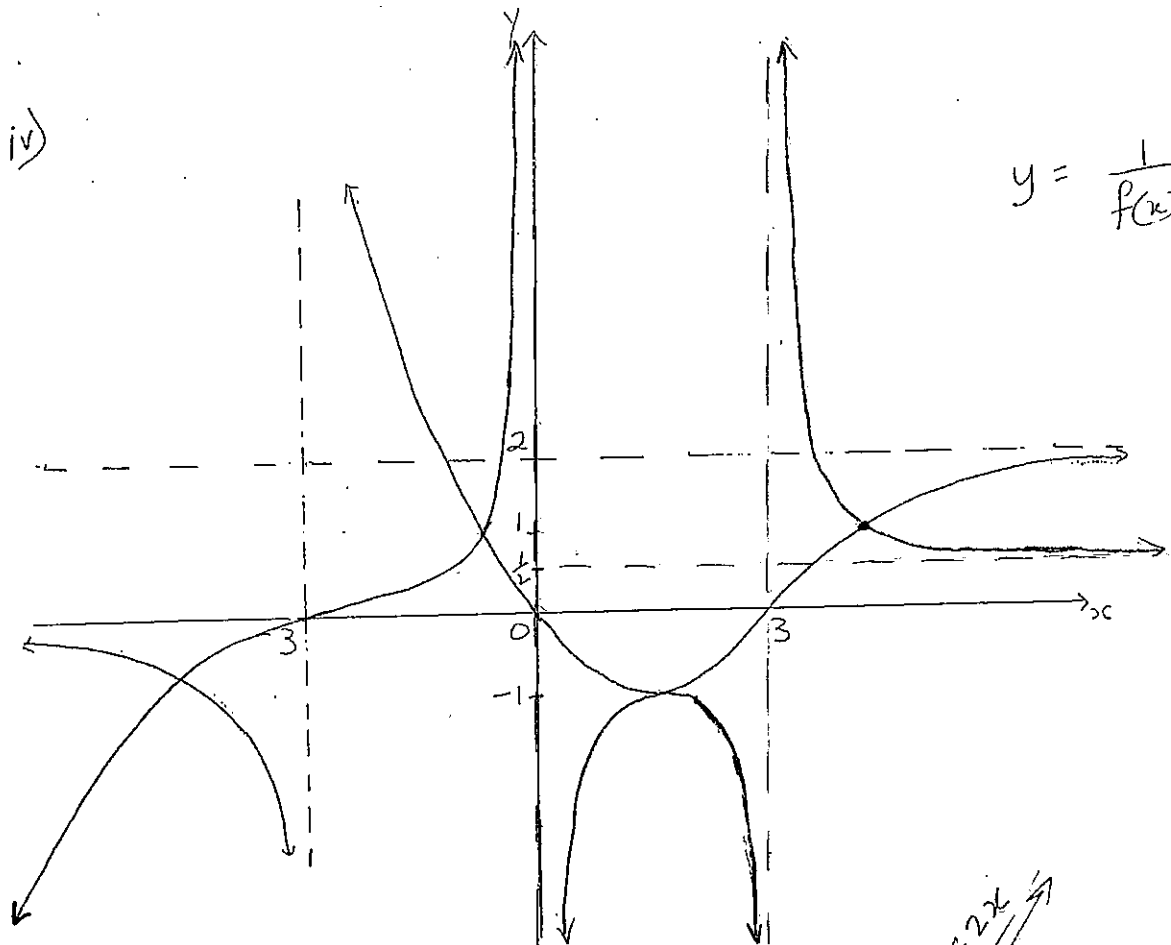
ii)

$$y^2 = f(x)$$



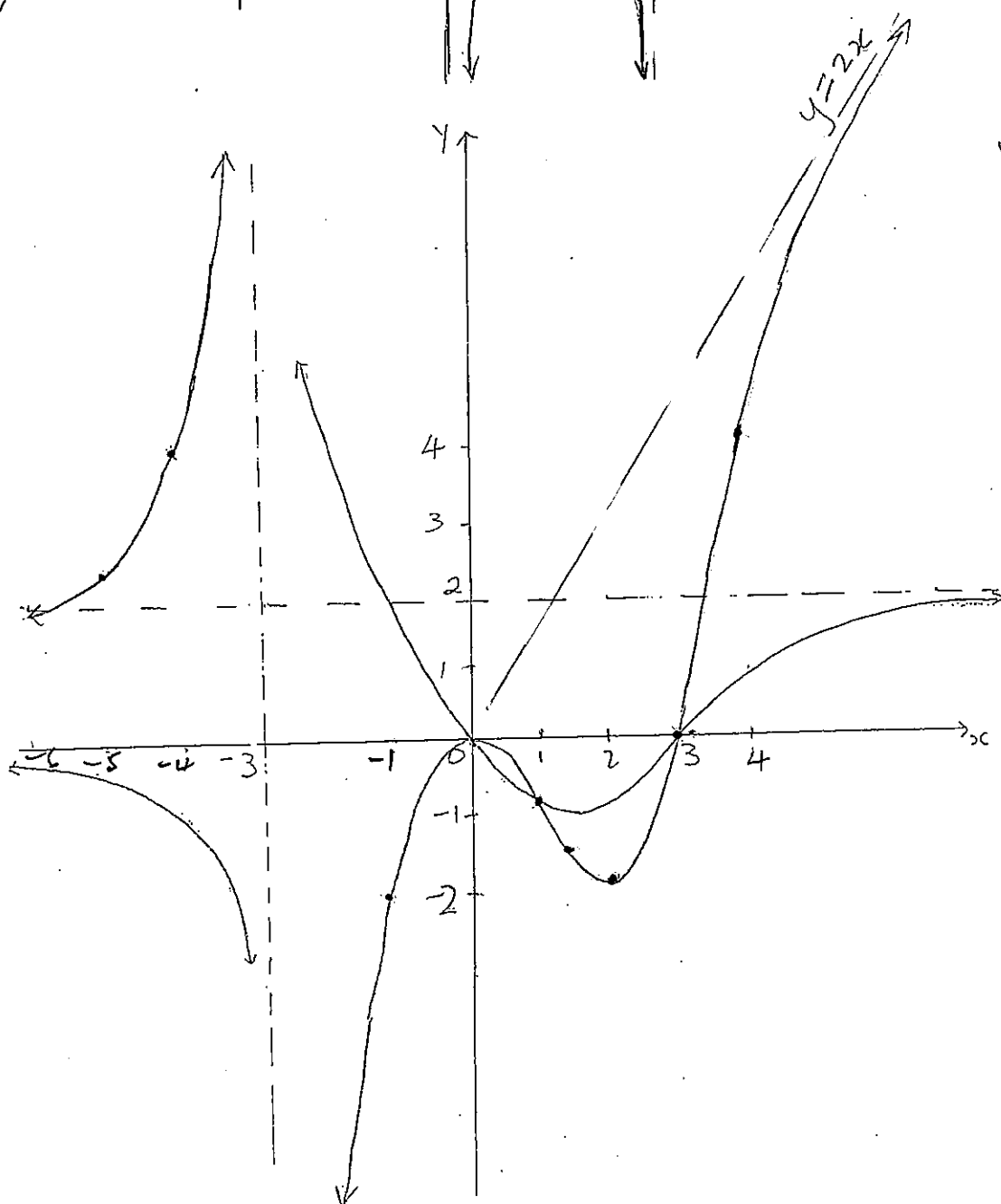
iv)

$$y = \frac{1}{f(x)}$$



v)

$$y = x f(x)$$



### Question 3b

i)  $xy = c^2$   
 $x \frac{dy}{dx} + y \cdot 1 = 0$  (Differentiating implicitly)

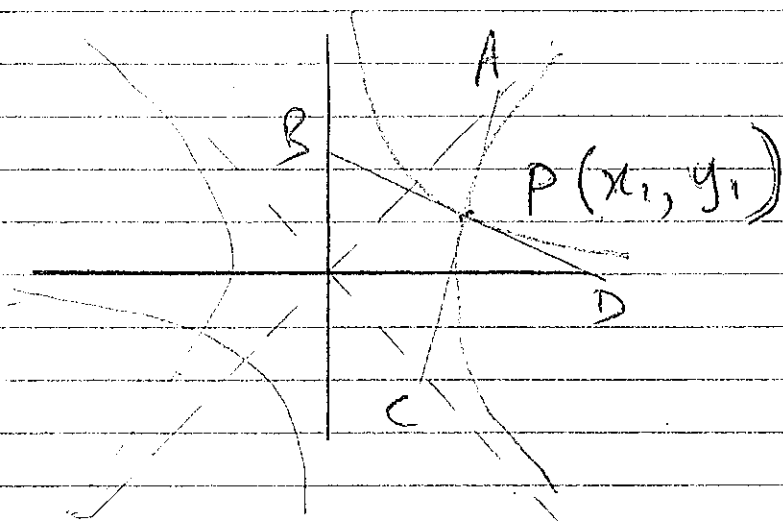
$$\therefore \frac{dy}{dx} = -\frac{y}{x} = \boxed{-\frac{y_1}{x_1}} \text{ for } P(x_1, y_1)$$

$\therefore$  Tangent is  
 $y - y_1 = -\frac{y_1}{x_1} (x - x_1)$

$$x_1 y - x_1 y_1 = -y_1 x + x_1 y_1$$

$$x_1 y_1 + x_1 y = 2x_1 y_1 = 2c^2 \text{ as required.}$$

ii)



For  $x^2 - y^2 = a^2$   
 $2x - 2y \frac{dy}{dx} = 0$

$$\frac{dy}{dx} = \frac{2x}{2y} = \frac{x}{y}$$

$\therefore$  at P,  $\frac{dy}{dx} = \boxed{\frac{x_1}{y_1}}$

Tangent is

$$y - y_1 = \frac{x_1}{y_1} (x - x_1)$$

$$y_1 y - y_1^2 = x_1 x - x_1^2$$

$$x_1^2 - y_1^2 = x_1 x - y_1 y$$

$$a^2 = x_1 x - y_1 y$$

This meets  $y = x$  at A:  $a^2 = x(x_1 - y_1)$

A is  $\left( \frac{a^2}{x_1 - y_1}, \frac{a^2}{x_1 - y_1} \right)$

this meets  $y = -x$  at C:  $a^2 = x(x_1 + y_1)$

C is  $\left( \frac{a^2}{x_1 + y_1}, -\frac{a^2}{x_1 + y_1} \right)$

Tangent  $xy_1 + x_1y = 2c^2$  meets  $y$  axis at B

$$B: \left(0, \frac{2c^2}{x_1}\right) = (0, 2y_1) \text{ as } y_1 = \frac{c^2}{x_1}$$

and  $x$  axis at D

$$D: \left(\frac{2c^2}{y_1}, 0\right) = (2x_1, 0) \text{ as } \frac{c^2}{y_1} = x_1$$

Now Mid Point of BD =  $\left(\frac{0+2x_1}{2}, \frac{2y_1+0}{2}\right)$   
=  $(x_1, y_1)$  i.e. P.

Mid Point of AC

$$x = \frac{\frac{a^2}{x_1 - y_1} + \frac{a^2}{x_1 + y_1}}{2}$$

$$y = \frac{\frac{a^2}{x_1 - y_1} + \frac{-a^2}{x_1 + y_1}}{2}$$

$$= \frac{a^2(x_1 + y_1 + x_1 - y_1)}{2(x_1 - y_1)(x_1 + y_1)}$$

$$y = \frac{a^2(x_1 + y_1 - x_1 + y_1)}{2(x_1 - y_1)(x_1 + y_1)}$$

$$= \frac{2a^2x_1}{2(x_1^2 - y_1^2)}$$

$$= \frac{2a^2y_1}{2(x_1^2 - y_1^2)}$$

But  $x_1^2 - y_1^2 = a^2$

$$\therefore x = \frac{2a^2x_1}{2a^2}$$

$$y = \frac{2a^2y_1}{2a^2}$$

$$x = x_1$$

$$y = y_1$$

$\therefore$  Mid Point of AC is  $(x_1, y_1)$  i.e. P.

$\therefore$  Diagonals bisect each other  $\therefore$   
ABCD is a parallelogram.

But Gradients of tangents  $m_1 \times m_2 = -\frac{y_1}{x_1} \times \frac{x_1}{y_1} = -1$

$\therefore$  Diagonals are perpendicular.

$\therefore$  ABCD is a rhombus.

Now length of BD :

$$d^2 = 4x_1^2 + 4y_1^2$$

$$d = 2\sqrt{x_1^2 + y_1^2}$$

Length of AC :

$$d^2 = \left( \frac{a^2}{x_1 + y_1} - \frac{a^2}{x_1 - y_1} \right)^2 + \left( \frac{-a^2}{x_1 + y_1} - \frac{a^2}{x_1 - y_1} \right)^2$$

$$= \left\{ \frac{a^2(x_1 - y_1 - x_1 - y_1)}{(x_1 + y_1)(x_1 - y_1)} \right\}^2 + \left\{ \frac{-a^2(x_1 - y_1 + x_1 + y_1)}{(x_1 + y_1)(x_1 - y_1)} \right\}^2$$

$$= \left\{ \frac{a^2(-2y_1)}{x_1^2 - y_1^2} \right\}^2 + \left\{ \frac{-a^2(2x_1)}{x_1^2 - y_1^2} \right\}^2$$

$$= \left\{ \frac{a^2(-2y_1)}{a^2} \right\}^2 + \left\{ \frac{-a^2(2x_1)}{a^2} \right\}^2$$

$$= 4y_1^2 + 4x_1^2$$

$$d = 2\sqrt{x_1^2 + y_1^2}$$

$\therefore$  BD = AC  $\therefore$  equal diagonals.

ABCD is now a square

#### Question 4

a) i)  $x = r \cos \theta$

$$\dot{x} = \frac{dx}{dt} = -r \sin \theta \frac{d\theta}{dt}$$

$$\dot{x} = -r\omega \sin \theta$$

as  $\theta = \omega t$  &  $\frac{d\theta}{dt} = \omega$

$$\ddot{x} = -r\omega \cos \theta \frac{d\theta}{dt}$$

$$\ddot{x} = -r\omega^2 \cos \theta$$

$$y = r \sin \theta$$

$$\dot{y} = \frac{dy}{dt} = r \cos \theta \frac{d\theta}{dt}$$

$$\dot{y} = r\omega \cos \theta$$

$$\ddot{y} = -r\omega \sin \theta \frac{d\theta}{dt}$$

as  $\omega$  is constant

$$\ddot{y} = -r\omega^2 \sin \theta$$

$$Acc = \sqrt{\ddot{x}^2 + \ddot{y}^2} = \sqrt{r^2 \omega^4 \cos^2 \theta + r^2 \omega^4 \sin^2 \theta}$$

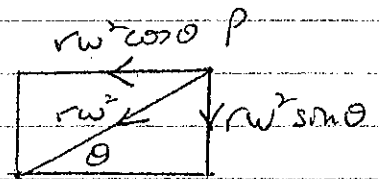
$$a = r\omega^2$$

Now  $F = ma$   $\therefore \underline{\underline{F = mr\omega^2}}$

Direction of this force

OP is resultant of  $\ddot{x}$  &  $\ddot{y}$

force is directed to centre of the circle



ii) Gravitational force  $F = \frac{Am}{r^2}$

As  $\omega$  is constant, this is equal to radial force  $mr\omega^2$ , (All other forces negligible)

$$mr\omega^2 = \frac{Am}{r^2}$$

$$r^3 = \frac{A}{\omega^2}$$

$$\underline{\underline{r = \sqrt[3]{\frac{A}{\omega^2}}}}$$

4b) i)  $(x-y)^2 \geq 0$  equality holds when  $x=y$

ie  $x^2 - 2xy + y^2 \geq 0$   
 $x^2 + y^2 \geq 2xy$  — (1)

ii)  $y^2 + z^2 \geq 2yz$  — (2)

$z^2 + x^2 \geq 2zx$  — (3)

(1)+(2)+(3)  $x^2 + y^2 + y^2 + z^2 + z^2 + x^2 \geq 2xy + 2yz + 2zx$

$x^2 + y^2 + z^2 \geq xy + yz + zx$

or  $x^2 + y^2 + z^2 - xy - yz - zx \geq 0$  — (4)

iii) (4)  $\times (x+y+z)$

$x^3 + xy^2 + xz^2 - x^2y - xyz - zx^2$   
 $+ x^2y + y^3 + yz^2 - xy^2 - y^2z - yxz$   
 $+ zx^2 + zy^2 + z^3 - xyz - yz^2 - z^2x \geq 0$

$x^3 + y^3 + z^3 - xyz - xyz - xyz \geq 0$

$x^3 + y^3 + z^3 \geq 3xyz$  — (5)

iv) Put  $x = x^{1/3}$ ,  $y = y^{1/3}$ ,  $z = z^{1/3}$

(5) becomes  $x + y + z \geq 3 x^{1/3} y^{1/3} z^{1/3}$  — (6)

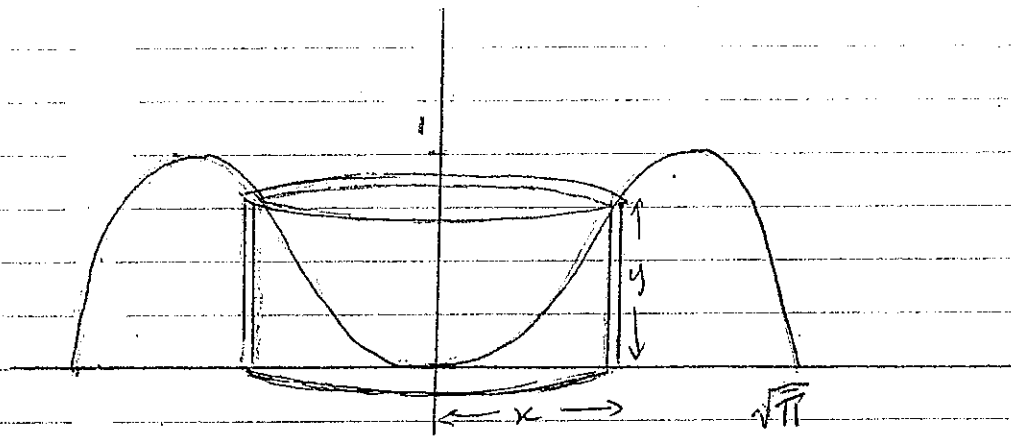
Put  $x = x^{-1/3}$ ,  $y = y^{-1/3}$ ,  $z = z^{-1/3}$  in (5)

(5) becomes  $x^{-1} + y^{-1} + z^{-1} \geq 3 x^{1/3} y^{1/3} z^{1/3}$  — (7)

(6)  $\times$  (7)  $(x+y+z)(x^{-1}+y^{-1}+z^{-1}) \geq 9 x^{1/3} x^{1/3} y^{1/3} y^{1/3} z^{1/3} z^{1/3}$

$(x+y+z)(x^{-1}+y^{-1}+z^{-1}) \geq 9$

4c)



$$\Delta V = 2\pi r h \Delta x$$

$$\Delta V = 2\pi x y \Delta x$$

$$\Delta V = 2\pi x \sin(x^2) \Delta x$$

$$V = \pi \int_0^{\sqrt{\pi}} 2x \sin(x^2) dx$$

$$= \pi \left[ -\cos(x^2) \right]_0^{\sqrt{\pi}}$$

$$= \pi (-\cos \pi + \cos 0)$$

$$= \pi (1 + 1)$$

$$V = 2\pi u^3$$

### Question 5

a) Let  $P(x) = (x-a)R(x)$   
+  $Q(x) = (x-a)T(x)$

$$P(x) - Q(x) = (x-a)R(x) - (x-a)T(x) \\ = (x-a)(R(x) - T(x))$$

$(x-a)$  is a factor of  $P(x) - Q(x)$

$$P(x) = x^3 + x^2 - 5x + k \\ Q(x) = x^3 - 8x^2 + 13x - 2k$$

have a common double factor  $(x-1)^2$

$\therefore P(x) - Q(x)$  has a factor  $(x-1)^2$

$$P(x) - Q(x) = x^3 + x^2 - 5x + k - (x^3 - 8x^2 + 13x - 2k) \\ = x^3 + x^2 - 5x + k - x^3 + 8x^2 - 13x + 2k \\ = 9x^2 - 18x + 3k$$

For a double root of a quadratic  $\Delta = 0$

$$18^2 - 4 \times 9 \times 3k = 0$$

$$2^2 \cdot 9^2 - 2^2 \cdot 9 \cdot 3k = 0$$

$$3k = 9$$

$$k = 3$$

$$\therefore P(x) - Q(x) = 9x^2 - 18x + 9 \\ = 9(x-1)^2$$

Double root is  $x=1$

2) i) Consider  $z = (\cos \theta + i \sin \theta)^5 = 1^5 \text{ cis } 5\theta$  By De Moivre's theorem

$$\therefore \text{cis } 5\theta = \cos^5 \theta + 5 \cos^4 \theta i \sin \theta - 10 \cos^3 \theta \sin^2 \theta - 10 i \cos^2 \theta \sin^3 \theta + 5 \cos \theta \sin^4 \theta + i \sin^5 \theta \quad (\text{Binomial theorem})$$

Equating real + imaginary parts:

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

$$+ \sin 5\theta = \sin^5 \theta - 10 \cos^2 \theta \sin^3 \theta + 5 \cos^4 \theta \sin \theta$$

$$\therefore \tan 5\theta = \frac{\sin^5 \theta - 10 \cos^2 \theta \sin^3 \theta + 5 \cos^4 \theta \sin \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta}$$

Dividing top + bottom by  $\cos^5 \theta$  (provided

$$\cos \theta \neq 0 \quad \text{ie } \theta \neq (2n-1)\frac{\pi}{2}$$

$$\tan 5\theta = \frac{\tan^5 \theta - 10 \tan^3 \theta + 5 \tan \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta} \quad \text{as required.}$$

iii) Put  $\tan 5\theta = 1$  and  $t = \tan \theta$

$$1 = \frac{t^5 - 10t^3 + 5t}{1 - 10t^2 + 5t^4}$$

$$1 - 10t^2 + 5t^4 = t^5 - 10t^3 + 5t$$

$$t^5 - 5t^4 - 10t^3 + 10t^2 + 5t - 1 = 0 \quad \text{has solutions equal to solutions of } \tan 5\theta = 1$$

$$\tan 5\theta = 1$$

$$5\theta = \frac{\pi}{4} + n\pi = (4n+1)\frac{\pi}{4}$$

$$\theta = (4n+1)\frac{\pi}{20}$$

Principal values:  $n=0 \Rightarrow \theta = \frac{\pi}{20}$  ;  $n=1 \Rightarrow \theta = \frac{5\pi}{20} = \frac{\pi}{4}$

$$n=2 \Rightarrow \theta = \frac{9\pi}{20} ; n=3 \Rightarrow \theta = \frac{13\pi}{20} ; n=4 \Rightarrow \theta = \frac{17\pi}{20}$$

$\therefore$  Solutions are

$$t = \tan \frac{\pi}{20}, \tan \frac{\pi}{4}, \tan \frac{9\pi}{20}, \tan \frac{13\pi}{20}, \tan \frac{17\pi}{20}$$

$\therefore$   $t = \tan \frac{\pi}{20}, \tan \frac{\pi}{4}, \tan \frac{9\pi}{20}, \tan \frac{13\pi}{20}, \tan \frac{17\pi}{20}$

would be the same values for  $t$ .

## Question 6

a) Taking Downwards as positive

$$m \ddot{x} = mg - mkv$$

$$\ddot{x} = g - kv$$

$$\frac{dv}{dt} = g - kv$$

$$\frac{dt}{dv} = \frac{1}{g - kv}$$

$$t = -\frac{1}{k} \ln(g - kv) + C_1$$

When  $t = 0$ ,  $v = 0$   $\therefore C_1 = \frac{1}{k} \ln g$

$$\therefore t = -\frac{1}{k} \ln \frac{g - kv}{g}$$

$$\text{or } -kt = \ln \frac{g - kv}{g}$$

$$e^{-kt} = \frac{g - kv}{g}$$

$$\underline{\underline{v = \frac{g}{k} (1 - e^{-kt})}}$$

ii) Taking upwards as positive

$$m \ddot{x} = -mg - mkv$$

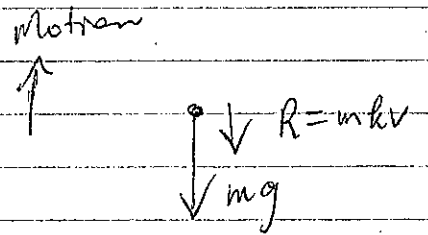
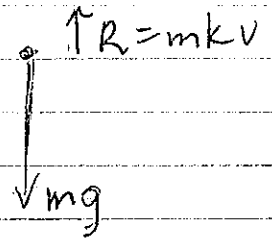
$$\ddot{x} = -g - kv$$

$$\frac{dt}{dv} = \frac{-1}{g + kv}$$

$$t = -\frac{1}{k} \ln(g + kv) + C_2$$

When  $t = 0$ ,  $v = u \Rightarrow C_2 = \frac{1}{k} \ln(g + ku)$

$$\text{or } t = -\frac{1}{k} \ln \left( \frac{g + kv}{g + ku} \right)$$



When 2nd particle is at rest  $v=0$

$$t = -\frac{1}{k} \ln\left(\frac{g}{g+ku}\right)$$

Velocity of 1st particle at this time

$$v = \frac{g}{k} (1 - e^{-kt})$$

$$v = \frac{g}{k} \left(1 - e^{\ln\left(\frac{g}{g+ku}\right)}\right)$$

$$v = \frac{g}{k} \left(1 - \frac{g}{g+ku}\right)$$

$$= \frac{g}{k} \left(\frac{g+ku - g}{g+ku}\right)$$

$$v = \frac{g}{k} \cdot \frac{ku}{g+ku} = \frac{gu}{g+ku} \quad *$$

Now terminal velocity of first particle is

$$\lim_{t \rightarrow \infty} \frac{g}{k} (1 - e^{-kt})$$

$$V = \frac{g}{k}$$

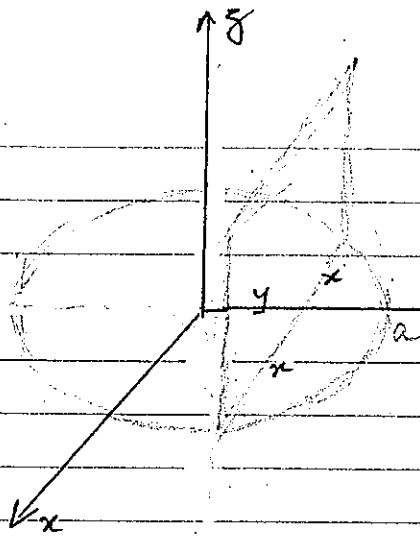
$\therefore *$  becomes

$$v = \frac{\frac{g}{k} u}{\frac{g}{k} + u}$$

$$v = \frac{Vu}{V+u}$$

as required

(6b)

At distance  $y$  from  $O$ 

$$x = \sqrt{a^2 - y^2}$$

Side of square is

$$2x = 2\sqrt{a^2 - y^2}$$

 $\therefore$  Area of Cross-section

$$A = (2\sqrt{a^2 - y^2})^2$$

$$= 4(a^2 - y^2)$$

$$\Delta V = 4(a^2 - y^2) \Delta y$$

$$V = \int_{-a}^a 4(a^2 - y^2) dy$$

$$= 2 \int_0^a 4(a^2 - y^2) dy$$

$$= 8 \left[ a^2 y - \frac{y^3}{3} \right]_0^a$$

$$V = 8 \left( a^3 - \frac{a^3}{3} \right)$$

$$V = \frac{16a^3}{3}$$

$$c) \quad \cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\cos(\alpha + \beta) + \cos(\alpha - \beta) = 2 \cos \alpha \cos \beta$$

$$\text{or } \cos \alpha \cos \beta = \frac{1}{2} \{ \cos(\alpha + \beta) + \cos(\alpha - \beta) \}$$

$$\cos mx \cos nx = \frac{1}{2} \{ \cos(m+n)x + \cos(m-n)x \}$$

$$\int_0^\pi \cos mx \cos nx dx = \frac{1}{2} \left[ \frac{\sin(m+n)x}{m+n} + \frac{\sin(m-n)x}{m-n} \right]_0^\pi$$

$$= \frac{1}{2} \frac{\sin(m+n)\pi}{m+n} + \frac{1}{2} \frac{\sin(m-n)\pi}{m-n} = 0$$

But  $\sin k\pi = 0$ 

$$\therefore I = 0$$

### Question 7

a) i) For P,  $\cos x = \tan x \implies \cos^2 x = \sin x$   
also  $y = \cos x$   $y = \tan x$   $\cos^2 x = \sin x$   
 $y' = -\sin x$   $y' = \sec^2 x$   
 $m_1 = -\sin x$   $m_2 = \sec^2 x$

$$m_1 \cdot m_2 = -\sin x \cdot \sec^2 x \\ = -\sin x \cdot \frac{1}{\cos^2 x}$$

But at P,  $\cos^2 x = \sin x$

$\therefore m_1 \cdot m_2 = -1$  as required.  
curves intersect at right angles at P.

ii) Now solving  $\cos^2 x = \sin x$

$$1 - \sin^2 x = \sin x$$

$$\sin^2 x + \sin x - 1 = 0$$

$$\sin x = \frac{-1 \pm \sqrt{1+4}}{2}$$

$$\sin x = \frac{-1 \pm \sqrt{5}}{2}$$

But  $\cos^2 x = \sin x$   $\therefore$  positive solution required  
 $\sin x = \frac{-1 + \sqrt{5}}{2} = \frac{\sqrt{5}-1}{2}$

$$\cos^2 x = \frac{\sqrt{5}-1}{2}$$

$$\sec^2 x = \frac{2}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1}$$

$$\sec^2 x = \frac{2(\sqrt{5}+1)}{5-1}$$

$$\sec^2 x = \frac{\sqrt{5}+1}{2} \text{ as required}$$

$$7b) \quad U_n = \int_0^1 (1-x^2)^n dx$$

Put  $u = (1-x^2)^n$   $dv = dx$   
 $du = n(1-x^2)^{n-1} \cdot -2x dx$   $v = x$

$$\begin{aligned} \int u dv &= uv - \int v du \\ \int_0^1 (1-x^2)^n dx &= \left[ x(1-x^2)^n \right]_0^1 - \int_0^1 nx \cdot 2x(1-x^2)^{n-1} dx \\ &= 0 - 2n \int_0^1 x^2 (1-x^2)^{n-1} dx. \end{aligned}$$

$$= -2n \int_0^1 (1-x^2-1)(1-x^2)^{n-1} dx$$

$$= -2n \left\{ \int_0^1 (1-x^2)^n dx - \int_0^1 (1-x^2)^{n-1} dx \right\}$$

$$U_n = -2n \{ U_n - U_{n-1} \}$$

$$= -2n U_n + 2n U_{n-1}$$

$$(2n+1) U_n = 2n U_{n-1}$$

$$U_n = \frac{2n}{2n+1} U_{n-1} \quad \text{as required}$$

$$\therefore U_4 = \frac{8}{9} U_3 \quad ; \quad U_3 = \frac{6}{7} U_2$$

$$U_2 = \frac{4}{5} U_1 \quad ; \quad U_1 = \frac{2}{3} U_0$$

$$U_0 = \int_0^1 (1-x^2)^0 dx = \int_0^1 1 dx = \left[ x \right]_0^1 = 1$$

$$\therefore U_4 = \frac{8}{9} \cdot \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} \cdot 1$$

$$U_4 = \frac{128}{315}$$

7c) i)  $\angle ALP + \angle AMP = 90^\circ + 90^\circ = 180^\circ$

Opposite angles are supplementary.

ii)  $\angle PMB = \angle PNB$  (given  $90^\circ$  each)  
 Equal angles subtended by interval PB  
 at points M & N (ie angles in same  
 segment standing on arc PB)

iii) Let  $\angle PBN = \alpha$   
 $\therefore \angle PMN = 180^\circ - \alpha$   
 (Opposite angle of cyclic quad PMNB)

Also  $\angle PAC = 180^\circ - \alpha$   
 (Opposite angles of cyclic quad PACB)

$\therefore \angle LAP = \alpha$  ( $\angle AC$  is a straight angle)

But  $\angle LAP = \angle LMP$  (Angles in same  
 segment standing on arc PL of cyclic  
 quad PLAM)

ie  $\angle LMP = \alpha$

$$\therefore \angle LMP + \angle PMN = \alpha + 180 - \alpha = 180^\circ$$

ie

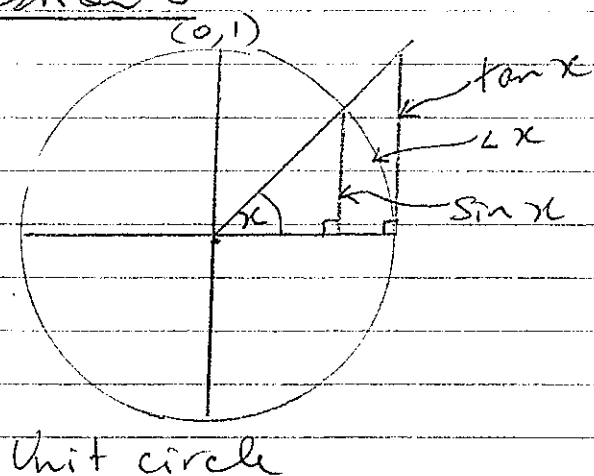
$\angle LMN$  is a straight angle

or

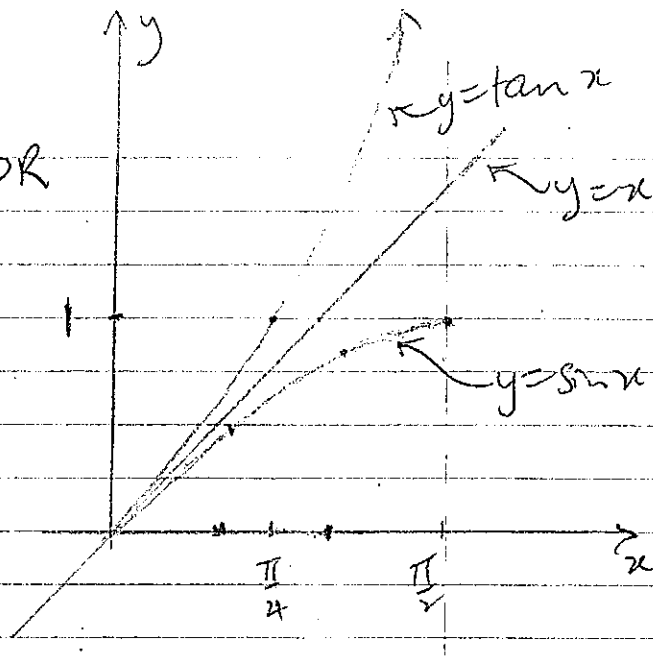
L, M & N are collinear.

# Question 8

a)



OR



As all are positive for  $0 < x < \frac{\pi}{2}$

area under  $\sin x < \text{area under } x < \text{area under } \tan x$

$x^2$  is also positive

$$\therefore x^2 \sin x < x^3 < x^2 \tan x$$

\* Areas under these are all positive

$$\therefore \int_0^{\pi/6} x^2 \sin x \, dx < \int_0^{\pi/6} x^3 \, dx < \int_0^{\pi/6} x^2 \tan x \, dx$$

$$\text{But } \int_0^{\pi/6} x^3 \, dx = \left[ \frac{x^4}{4} \right]_0^{\pi/6} = \frac{\pi^4}{4 \times 6^4} = \frac{\pi^4}{2^6 \cdot 3^4}$$

$$\therefore \int_0^{\pi/6} x^2 \sin x \, dx < \frac{\pi^4}{2^6 \cdot 3^4} < \int_0^{\pi/6} x^2 \tan x \, dx$$

$$\text{b) Put } \alpha = \tan^{-1} 3x \Rightarrow \tan \alpha = 3x$$

$$\therefore \beta = \tan^{-1} 2x \Rightarrow \tan \beta = 2x$$

$$\therefore (\alpha - \beta) = \tan^{-1} \frac{1}{5}$$

$$\text{or } \tan(\alpha - \beta) = \frac{1}{5}$$

$$\frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} = \frac{1}{5}$$

$$\frac{3x - 2x}{1 + 6x^2} = \frac{1}{5}$$

$$5x = 1 + 6x^2$$

$$6x^2 - 5x + 1 = 0$$

$$(3x-1)(2x-1) = 0$$

$$x = \frac{1}{3} \quad \text{or} \quad \frac{1}{2}$$

c) i)  $\ddot{x} = 0$   $\ddot{y} = -g$   
 $\dot{x} = c_1$   $\dot{y} = -gt + c_2$   
 $t=0, \dot{x} = V \cos \theta, \dot{y} = V \sin \theta$   
 $\therefore \dot{x} = V \cos \theta$   $\dot{y} = -gt + V \sin \theta$   
 $x = Vt \cos \theta + c_3$   $y = -\frac{gt^2}{2} + Vt \sin \theta + c_4$   
 $t=0, x=0, y=0$   
 $\therefore x = Vt \cos \theta$   $y = -\frac{gt^2}{2} + Vt \sin \theta$

ii)  $t = \frac{x}{V \cos \theta}$   
 $\therefore y = \frac{-gx^2}{2V^2 \cos^2 \theta} + V \cdot \frac{x}{V \cos \theta} \sin \theta$

$$y = \frac{-gx^2 \sec^2 \theta}{2V^2} + x \tan \theta$$

iii) Inclined plane:  $y = x \tan \alpha$   
 $\therefore x \tan \alpha = \frac{-gx^2 \sec^2 \theta}{2V^2} + x \tan \theta$

$$g \sec^2 \theta x^2 + 2V^2 x \tan \alpha - 2V^2 x \tan \theta = 0$$

$$x (g \sec^2 \theta x + 2V^2 (\tan \alpha - \tan \theta)) = 0$$

$$x = 0 \quad \text{or} \quad x = \frac{2V^2 (\tan \theta - \tan \alpha)}{g \sec^2 \theta}$$

$$\therefore R = \frac{2V^2}{g} (\tan \theta - \tan \alpha) \cos^2 \theta \sec \alpha$$

$$\text{or } R = \frac{2V^2}{g} (\sin \theta \cos \theta - \tan \alpha \cos^2 \theta) \sec \alpha$$

$$R = \frac{2V^2}{g} \left( \frac{1}{2} \sin 2\theta - \tan \alpha \cos^2 \theta \right) \sec \alpha$$

v) Now  $\frac{dR}{d\theta} = \frac{2V^2}{g} \left( \frac{1}{2} \cdot 2 \cos 2\theta - 2 \cos \theta (\sin \theta) \tan \alpha \right) \sec \alpha$

$$\frac{dR}{d\theta} = \frac{2V^2}{g} (\cos 2\theta + \sin 2\theta \tan \alpha) \sec \alpha$$

vi) For maximum range  $\frac{dR}{d\theta} = 0$ .

$$\cos 2\theta + \sin 2\theta \tan \alpha = 0$$

$$\sin 2\theta \tan \alpha = -\cos 2\theta$$

$$\tan 2\theta = -\frac{1}{\tan \alpha}$$

$$\tan 2\theta = -\cot \alpha$$

$$\tan 2\theta = -\tan \left( \frac{\pi}{2} - \alpha \right)$$

$$\therefore 2\theta = \pi - \left( \frac{\pi}{2} - \alpha \right) \quad \text{or} \quad 2\pi - \left( \frac{\pi}{2} - \alpha \right)$$

$$2\theta = \frac{\pi}{2} + \alpha$$

$$\text{or } \frac{3\pi}{2} + \alpha$$

$$\theta = \frac{\pi}{4} + \frac{\alpha}{2}$$

$$\text{or } \frac{3\pi}{4} + \frac{\alpha}{2}$$

$$\text{But } 0 < \theta < \frac{\pi}{2}$$

$$\& \quad \theta > \alpha$$

$$\therefore \theta = \frac{\pi}{4} + \frac{\alpha}{2}$$

(Minimum range is obviously when  $\theta = \frac{\pi}{2}$ ,  $R=0$ )